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METHOD FOR CONTROLLING POWER CONVERTER AND POWER CONVERTER

Abstract

The present disclosure provides a method for controlling a power converter and a power converter. The power converter includes a first switch, a second switch, a first driving unit corresponding to the first switch and a second driving unit corresponding to the second switch, and a bootstrap capacitor configured to supply power to the second driving unit. When the first switch is turned on, the bootstrap capacitor is charged via the first switch. The method includes: when a duty cycle of the first switch is decreased to a first threshold, controlling the duty cycle of the first switch to be a preset duty cycle every time at least one switching cycle elapses, and the preset duty cycle is not less than the first threshold.

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Background/Summary

CROSS REFERENCE

[0001] The present application is based on and claims priority to Chinese Patent Application No. 202410238672X, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of power supply technologies, and in particular to a method for controlling a power converter and a power converter.

BACKGROUND

[0003] For a power converter whose switching tube is driven by bootstrap driving, for example, as shown in FIGS. 1 and 2, a bootstrap capacitor **205** of an upper switching tube (a second switch **202**) needs to be charged by turning on a lower switching tube (a first switch **201**) during normal operation. When some special operating conditions, such as no-load, occur, the lower switching tube of the power converter will enter a drive-off state. In this case, the lower switching tube is turned off, a voltage of the bootstrap capacitor **205** of the upper switching tube cannot be effectively supplemented, and a voltage across the bootstrap capacitor will continue to drop. As a result, when the driving of the lower switching tube is restored and the lower switching tube is turned on, the upper switching tube will generate a voltage jump and this voltage jump cannot be absorbed, thereby causing the upper switching tube and the lower switching tube to be turned on simultaneously, so that the circuit is at risk of damage.

[0004] It should be noted that the information disclosed in the Background section above is only for enhancing the understanding of the background of the present disclosure, and thus may include information that does not constitute prior art known to those of ordinary skill in the art.

SUMMARY

[0005] The present disclosure provides a method for controlling a power converter and a power converter.

[0006] According to a first aspect of the present disclosure, there is provided a method for controlling a power converter,

[0007] the power converter includes a first switch, a second switch, a first driving unit corresponding to the first switch and a second driving unit corresponding to the second switch, and a bootstrap capacitor configured to supply power to the second driving unit, when the first switch is turned on, the bootstrap capacitor is charged via the first switch, wherein the method includes:

[0008] when a duty cycle of the first switch is decreased to a first threshold, controlling the duty cycle of the first switch to be a preset duty cycle every time at least one switching cycle elapses, wherein the preset duty cycle is not less than the first threshold.

[0009] According to a second aspect of the present disclosure, there is further provided a power converter, including:

[0010] a first switch and a second switch, electrically connected to form a common node;

[0011] a first driving unit, powered by a driving power supply and configured to drive the first switch;

[0012] a second driving unit, powered by a bootstrap capacitor and configured to drive the second switch; and

[0013] a control unit, electrically connected to the first driving unit and the second driving unit, and configured to:

[0014] when a duty cycle of the first switch is decreased to a first threshold, control the duty cycle of the first switch to be a preset duty cycle every time at least one switching cycle elapses, wherein the preset duty cycle is not less than the first threshold.

[0015] It should be noted that the above general description and the following detailed description are merely exemplary and explanatory and should not be construed as limiting of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The drawings herein are incorporated in and constitute a part of the specification, illustrate embodiments consistent with the present disclosure, and together with the description serve to explain principles of the present disclosure. Apparently, the drawings in the following description are only some embodiments of the present disclosure. For those of ordinary skill in the art, other drawings may be obtained based on these drawings without paying any creative effort.

[0017] FIG. 1 shows a schematic structural diagram of a Boost circuit in the prior art;

[0018] FIG. 2 shows a schematic structural diagram of a driving circuit of the Boost circuit shown in FIG. 1;

[0019] FIG. 3 shows a schematic structural diagram of a totem pole PFC circuit in the prior art;

[0020] FIG. 4 shows a schematic diagram of a method for controlling a power converter according to the present disclosure;

[0021] FIG. 5 shows a first simulation waveform diagram of a power converter according to the present disclosure;

[0022] FIG. 6 shows a second simulation waveform diagram of a power converter according to the present disclosure;

[0023] FIG. 7 shows another schematic structural diagram of a driving circuit of the Boost circuit shown in FIG. 1; and

[0024] FIG. 8 shows a schematic diagram of an external connection relationship of a power converter according to the present disclosure.

DETAILED DESCRIPTION

[0025] Example embodiments will now be described more fully with reference to the accompanying drawings. However, the example embodiments can be implemented in a variety of forms and should not be construed as being limited to examples set forth herein; rather, these embodiments are provided so that the present disclosure will be more complete and comprehensive so as to convey the idea of the example embodiments to those skilled in this art. The described features, structures or characteristics may be combined in any suitable manner in one or more embodiments.

[0026] In addition, the drawings are merely schematic representations of the present disclosure and are not necessarily drawn to scale. The same reference numerals in the drawings denote the same or similar parts, and the repeated description thereof will be omitted. Some of block diagrams shown in the figures are functional entities and do not necessarily correspond to physically or logically separate entities. These functional entities may be implemented in software, or implemented in one or more hardware modules or integrated circuits, or implemented in different networks and/or processor devices and/or microcontroller devices.

[0027] As described above, for a power converter whose switching tube is driven by bootstrap driving, for example, as shown in FIGS. 1 and 2, a bootstrap capacitor **205** of an upper switching tube (a second switch **202**) needs to be charged by turning on a lower switching tube (a first switch **201**) during normal operation. When some special operating conditions, such as no-load, occur, the

lower switching tube of the power converter will enter a drive-off state. In this case, the lower switching tube is turned off, a voltage of the bootstrap capacitor **205** of the upper switching tube cannot be effectively supplemented, and a voltage across the bootstrap capacitor will continue to drop. As a result, when the driving of the lower switching tube is restored and the lower switching tube is turned on, the upper switching tube will generate a voltage jump and this voltage jump cannot be absorbed, thereby causing the upper switching tube and the lower switching tube to be turned on simultaneously, so that the circuit is at risk of damage.

[0028] The present disclosure provides a method for controlling a power converter and a power converter, which at least to a certain extent solve a problem in the related arts that the voltage of the bootstrap capacitor of the upper switching tube (the second switch) continues to drop when some special operating conditions (such as no-load) occur. In addition, the present disclosure can also avoid overshoot of an inductor current of the power converter and prevent the output voltage from being too large, thereby protecting the circuit.

[0029] In the method for controlling the power converter provided by the present disclosure, the duty cycle of the first switch and the second switch is adjusted to maintain the stability of the voltage of the bootstrap capacitor of the second switch (the upper switching tube), thereby avoiding the situation where the second switch has a voltage jump and the voltage jump cannot be absorbed, thereby causing the first switch and the second switch to be turned on simultaneously and damaging the circuit. In addition, the present disclosure can also avoid the overshoot of the inductor current and prevent the output voltage from being too large.

[0030] Specific embodiments of the present disclosure are described in detail below in conjunction with the accompanying drawings.

[0031] As shown in FIG. **1**, when a power converter is a Boost circuit, the power converter includes:

[0032] an input terminal **Vin1** which includes a first input terminal **Vin1+** and a second input terminal **Vin1—**, and an output terminal **Vout1** which includes a first output terminal **Vout1+** and a second output terminal **Vout1—**; a first switch **201** and a second switch **202** which are electrically connected between the first output terminal **Vout1+** and the second output terminal **Vout1—**, where an electrical connection point of the first switch **201** and the second switch **202** forms a common node **203**, the first switch **201** is electrically connected between the common node **203** and the second output terminal **Vout1—**, and the second switch **202** is electrically connected between the first output terminal **Vout1+** and the common node **203**; and an inductor **L1**, electrically connected between the first input terminal **Vin1+** and the common node **203**. The second input terminal **Vin1—** and the second output terminal **Vout1—** are electrically connected to a ground terminal **204**.

[0033] As shown in FIG. **2**, a driving circuit of the Boost circuit shown in FIG. **1** includes:

[0034] a first driving unit **209**, a first terminal of which is electrically connected to a control terminal of the first switch **201**, a second terminal of which is electrically connected to the ground terminal **204**, and controlling the on and off of the first switch **201**;

[0035] a second driving unit **210**, a first terminal of which is electrically connected to a control terminal of the second switch **202**, a second terminal of which is electrically connected to the common node **203**, and controlling the on and off of the second switch **202**;

[0036] a bootstrap capacitor **205**, a first terminal of which is electrically connected to a third terminal of the second driving unit **210**, and a second terminal of which is electrically connected to the common node **203**;

[0037] a driving power supply **208**, a first terminal of which is electrically connected to a third terminal of the first driving unit **209** and an anode of a diode **207**, and a second terminal of which is electrically connected to the ground terminal **204**, a cathode of the diode **207** is electrically connected to the third terminal of the second driving unit **210**.

[0038] In this embodiment, the driving power supply **208** is a Direct Current (DC) power supply. The first switch **201** and the second switch **202** may be switching tubes, including but not limited

to Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) and Insulated Gate Bipolar Transistors (IGBTs). The first driving unit **209** and the second driving unit **210** include but are not limited to driving chips.

[0039] In this embodiment, a driving mode of the power converter is bootstrap driving. Since a reference ground of the first driving unit **209** and a reference ground of the driving power supply **208** are both the ground terminal **204**, the driving power supply **208** can directly supply power to the first driving unit **209**, and accordingly, a voltage across the first driving unit **209** is stable. As for the second driving unit **210**, since its reference ground is the common node **203** (Gnd_up) of the first switch **201** and the second switch **202**, which is a voltage jump point, the driving power supply **208** cannot directly supply power to the second driving unit **210**. The bootstrap capacitor **205** is required to supply power to the second driving unit **210**. However, when the bootstrap capacitor **205** is charged, the first switch **201** needs to be turned on to form a loop for charging the bootstrap capacitor **205**. That is, when the first switch **201** is turned on, the bootstrap capacitor **205**, the first switch **201**, the driving power supply **208** and the diode **207** form a charging loop to charge the bootstrap capacitor **205**.

[0040] Under normal operating conditions, the first switch **201** can achieve the charge and discharge balance of the bootstrap capacitor **205**. But in special cases where the driving of the first switch **201** needs to be turned off, such as no-load, the first switch **201** remains off, so that the charging loop is disconnected, and the voltage of the bootstrap capacitor **205** gradually decreases. As a result, the second driving unit **210** loses power and cannot absorb the current from both terminals (gate-source) of G and S of the second switch **202**. When the driving of the first switch **201** is re-enabled and turned on, a voltage jump occurs between the gate and the source of the second switch **202**. This voltage jump cannot be absorbed, which may cause the first switch **201** and the second switch **202** to be turned on simultaneously, so that the circuit is at risk of damage.

[0041] For the power converter shown in FIG. 1, when the circuit operates in a special condition and a duty cycle of the first switch **201** gradually decreases or even becomes zero, it is necessary to forcibly change the duty cycle of the first switch **201** to a non-zero value. This ensures the charge and discharge balance of the bootstrap capacitor **205** of the second switch **202**, allowing the circuit to operate normally. For all power converters, when the special operating condition such as a no-load state occurs or the load is switched and the duty cycle of the first switch is continuously reduced, the duty cycle of the first switch may be extremely small or even zero. In these cases, it is necessary to adopt a control method in the present disclosure to provide sufficient energy for the bootstrap capacitor. For a power converter whose input power supply is an Alternating Current (AC) power supply, such as a totem pole PFC circuit, the special operating conditions also include that when the input power supply of the circuit is powered off and has not yet been restored. Since a polarity of the input power supply after restoration is unknown, it is impossible to determine whether the first switch or the second switch is a main switching tube. Therefore, it is necessary to force the duty cycles of the first switch and the second switch to be reduced to zero. In this case, the control method disclosed herein is also required to provide sufficient energy for the bootstrap capacitor.

[0042] It should be noted that the power converter includes two switches, and the second switch (the upper switching tube) needs to be powered by the bootstrap capacitor for driving. The power converter may have a variety of specific forms, which can be set according to actual circuit requirements. For example, the power converter may include a Boost circuit, a totem pole Power Factor Correction (PFC) circuit or a three-phase PFC circuit.

[0043] When the power converter is the totem pole PFC circuit, FIG. 3 shows a schematic structural diagram of a totem pole PFC circuit in the prior art, which includes: a first switch **301**, a second switch **302**, an input power supply Vin2, an inductor L2, a first diode D11, a second diode D12 (either or both of the two diodes may be replaced by a switch), and an output terminal Vout2. A first terminal (a first input terminal) of the input power supply Vin2 is connected to a first

terminal of the inductor L2, and a second terminal of the inductor L2 is connected to a common node 303 of the first switch 301 and the second switch 302. A second terminal of the second switch 302 is connected to a cathode of the first diode D11, a first terminal of the first switch 301 is connected to an anode of the second diode D12, and a common node formed by a cathode of the second diode D12 and an anode of the first diode D11 is connected to a second terminal (a second input terminal) of the input power supply Vin2. The cathode of the first diode D11 is electrically connected to a first output terminal Vout2+ of the output terminal Vout2, and the anode of the second diode D12 is electrically connected to a second output terminal Vout2- of the output terminal Vout2.

[0044] Embodiments of the present disclosure provide a method for controlling a power converter, applicable to a Boost circuit or a PFC circuit in which both switches on one bridge arm are switching tubes. However, when the specific structure of the circuit is different, a turn-on sequence of the first switch and the second switch is different. As shown in FIG. 4, the control method in the present disclosure includes:

[0045] In S1, when a duty cycle of the first switch decreases to a first threshold, the duty cycle of the first switch is controlled to be a preset duty cycle every time at least one switching cycle elapses, and the preset duty cycle is not less than the first threshold.

[0046] For the step S1, when the power converter is in different states, a change direction (increase/decrease) of the duty cycle of the first switch (the lower switching tube) is different. As mentioned above, the premise of the present disclosure is that the power converter enters a special operating condition, thereby causing the duty cycle of the first switch to gradually decrease or even become zero. In this case, in order to prevent the bootstrap capacitor of the second switch from continuously losing power, the duty cycle of the first switch is forcibly changed to a preset duty cycle when the duty cycle of the first switch is reduced to the first threshold. Accordingly, the first switch does not remain off continuously. Instead, whenever the first switch is turned on, the driving power supply charges the bootstrap capacitor, thereby stabilizing the voltage of the bootstrap capacitor. In addition, if the driving power supply charges the bootstrap capacitor enough each time the first switch is turned on, the first switch may also be turned on once every several switching cycles, which also help stabilize the voltage of the bootstrap capacitor. An interval of the switching cycle can be set according to actual needs.

[0047] In some special operating conditions, when the first switch is used as a main switching tube, the duty cycle of the first switch should gradually become zero. However, the control method disclosed herein forces it to become the preset duty cycle. Although the voltage of the bootstrap capacitor of the second switch is stabilized, this also brings another problem, that is, an output voltage continues to increase, which also affects the operating condition of the power converter. As shown in FIG. 5, where the driving waveforms are all driving waveforms of the first switch (the lower switching tube). This is because: the inductor resonates with a parasitic capacitor of the first switch and a parasitic capacitor of the second switch. As a result, an inductor current continues to increase after the first switch is turned off and before a parasitic diode of the second switch is turned on. To solve this problem, it is necessary to turn on the second switch in time after the first switch is turned off and to forcibly adjust a duty cycle of the second switch. In some embodiments, for the convenience of design, the duty cycle of the second switch is also set to the preset duty cycle. This forcibly changes a current commutation path, allowing the inductor current to decrease in time and ending resonance in advance. This ensures that the output voltage does not become too large, thereby effectively protecting the circuit. As shown in FIG. 6, where the driving waveforms are driving waveforms of the first switch (the lower switching tube) and the second switch (the upper switching tube) in sequence. Therefore, the control method further includes:

[0048] in S2, when an input voltage of the power converter is positive and the duty cycle of the first switch is controlled to be the preset duty cycle, the second switch is controlled to be turned on after a dead time following the turn-off of the first switch.

[0049] In some embodiments, for the step S2, the duty cycle of the second switch is also the preset duty cycle.

[0050] In some embodiments, in order to avoid a short circuit caused by turning on two switches at the same time, the second switch may be turned on after a period of time has elapsed since the first switch was turned off. This period of time is called the dead time, and the specific duration of the dead time may be set according to actual needs.

[0051] In some embodiments, the polarity of the input voltage may be determined based on a type of the input power supply of the power converter. When the input power supply is a DC power supply, the input voltage is a DC voltage and is always considered positive. When the input power supply is an AC power supply, the input voltage of the power converter is positive during the positive half cycle of the AC power supply.

[0052] For the Boost circuit shown in FIG. 1, its input power supply V_{in1} is a DC power supply. No matter the inductor $L1$ is connected to the positive or negative electrode of the input power supply V_{in1} , the input voltage can be considered to be positive. This is because the first switch **201** functions as the main switching tube, while the second switch **202** always acts as a synchronous rectifier tube. For a conventional Boost circuit using a pulse width modulation, under normal circumstances, the duty cycle of the first switch **201** (the lower switching tube) will gradually decrease to zero when the Boost circuit is in a special operating condition, such as no-load. As mentioned above, if this is not changed, when the duty cycle of the first switch **201** is zero, the charging loop of the bootstrap capacitor **205** of the second switch **202** (the upper switching tube) is disconnected, causing the voltage of the bootstrap capacitor **205** to gradually decrease. In order to stabilize the voltage of the bootstrap capacitor **205** of the second switch **202**, the present disclosure forcibly changes the normal change process of the duty cycle of the first switch **201** when the duty cycle of the first switch **201** is reduced to the first threshold. The duty cycle of the first switch **201** is then adjusted to the preset duty cycle, thereby ensuring the charging of the bootstrap capacitor **205** of the second switch **202**. The preset duty cycle is a non-zero value, selected within a specific range. The maximum value and minimum value of the specific range will be described in detail below. In addition, since the first switch **201** is always the main switching tube, controlling the duty cycle of the first switch **201** to the preset duty cycle will cause the output voltage to increase continuously (as described above). Therefore, after the first switch **201** is turned on with the preset duty cycle, the second switch **202** needs to be turned on immediately after the dead time.

[0053] For the totem pole PFC circuit shown in FIG. 3, since the input power supply V_{in2} is an AC power supply, there are two directions of the input voltage, and an input voltage in the positive half cycle of the AC power supply is defined as positive. In this case, the first switch **301** is used as the main switching tube, and the second switch **302** is used as the synchronous rectifier tube. When the first switch **301** is turned on, the current path is: the input power supply V_{in2} .fwdarw.the inductor $L2$.fwdarw.the first switch **301**.fwdarw.the second diode $D12$.fwdarw.the input power supply V_{in2} . It should be understood that in the negative half cycle of the AC power supply, the second switch **302** is used as the main switching tube, and the first switch **301** is used as the synchronous rectifier tube. When the second switch **302** is turned on, the current path is: the input power supply V_{in2} .fwdarw.the first diode $D11$.fwdarw.the second switch **302**.fwdarw.the inductor $L2$.fwdarw.the input power supply V_{in2} . Therefore, when the input voltage is positive, that is, during the positive half cycle of the AC power supply, the problem of the output voltage continuously increasing due to the duty cycle of the first switch **301** being set to the preset duty cycle will occur. Therefore, it is necessary to turn on the second switch **302** immediately after the first switch **301** is turned on with the preset duty cycle. To address this, the second switch **302** is turned on after the dead time. In the negative half cycle of the AC power supply, the second switch **302** is the main switching tube, and the duty cycle of the first switch **301** is set to the preset duty cycle, which does not cause the output voltage to increase. Therefore, it is not necessary to turn on the second switch **302** immediately after the first switch **301** is turned on with the preset duty cycle.

[0054] In some embodiments, when the power converter is the Boost circuit, the totem pole PFC circuit or the three-phase PFC circuit, the duty cycle of the first switch gradually decreases to zero in a no-load state according to a conventional control strategy. In this case, when the duty cycle of the first switch is reduced to the first threshold, the change process of the duty cycle of the first switch is forcibly changed to make the duty cycle of the first switch become the preset duty cycle, thereby ensuring the charging of the bootstrap capacitor and stabilizing the voltage of the bootstrap capacitor. If the power converter is in a load switching state, for a case where the duty cycle of the first switch gradually decreases, there is a state in which the duty cycle of the first switch is reduced to the first threshold. In this case, if the change process of the duty cycle of the first switch is not forcibly changed, the duty cycle of the first switch may be reduced to below the first threshold (which is not necessarily zero). In this case, the charging of the bootstrap capacitor is not sufficient to drive the normal operation of the second switch, affecting the normal operation of the circuit. For this, it is also necessary to adopt the control method in the present disclosure. When the duty cycle of the first switch is reduced to the first threshold, the duty cycle of the first switch is forcibly changed to become the preset duty cycle, thereby stabilizing the voltage of the bootstrap capacitor and maintaining the normal operation of the circuit.

[0055] In some embodiments, when the power converter is the totem pole PFC circuit, if the input power of the circuit is powered off and has not yet been restored, the conventional control strategy requires reducing the duty cycle of the first switch and the second switch to be reduced to zero. Therefore, when the duty cycle of the first switch is reduced to the first threshold, in order to stabilize the voltage of the bootstrap capacitor, the duty cycle of the first switch is forcibly changed to the preset duty cycle.

[0056] In some embodiments, a selection range of the preset duty cycle has the minimum value and the maximum value. The minimum value of the preset duty cycle is defined as the first threshold. Specifically, the minimum value (the first threshold) of the preset duty cycle may be determined based on the sum of the energy required to drive the second driving unit and the energy required to turn on the second switch. That is to say, when the first switch is turned on according to the minimum value of the preset duty cycle, the energy provided by the driving power supply is greater than the sum of the energy required to drive the second driving unit and the energy required to turn on the second switch. In addition, as shown in FIG. 8, the maximum value of the preset duty cycle may be determined based on an input power when the input voltage of the power converter **801** is maximum and the power consumed by an auxiliary power supply **802** electrically connected to the output terminal of the power converter **801**. When the first switch is turned on with the maximum value of the preset duty cycle, and the input voltage of the power converter **801** is at its maximum, the input power of the power converter **801** is less than the power consumed by the auxiliary power supply **802**. This is because: in order to maintain the voltage stability of the bootstrap capacitor, the first switch needs to be turned on according to the preset duty cycle, but it also brings a new problem: when the input voltage of the power converter is relatively high, the output voltage will continue to increase if the first switch is turned on according to a larger preset duty cycle, thereby causing damage to the circuit. This is because the premise the present disclosure is that the power converter enters a special operating condition where the power consumed by the load of the power converter is small or close to zero. Since the auxiliary power supply is electrically connected to the output terminal of the power converter, the power output by the power converter is mainly used to supply energy to the auxiliary power supply. The energy consumed by the auxiliary power supply is used to maintain the normal operation of the circuit. Only when the input power is less than the power consumed by the auxiliary power supply, the output voltage will not continue to increase, thereby effectively protecting the circuit.

[0057] By determining the minimum and maximum values of the preset duty cycle, a value range of the preset duty cycle can be determined, and the value of the preset duty cycle can be selected within this value range. The selection of the specific value of the preset duty cycle can be

determined according to actual needs.

[0058] It can be seen that in the method for controlling the power converter provided by the present disclosure, it is only necessary to change the duty cycle of the first switch when the power converter enters the special operating condition, and turn on the second switch in time when the input voltage is positive, so as to maintain the stability of the voltage of the bootstrap capacitor that provides the driving to the second switch, avoiding the circuit damage caused by the direct conduction of the first switch and the second switch. In addition, the present disclosure can also avoid the overshoot of the inductor current and prevent the output voltage from being too large, thereby protecting the circuit. Compared with the structure in the prior art, there is no additional hardware structure, and the safety of the circuit can be greatly improved in a low-cost manner.

[0059] Based on the same inventive concept, the present disclosure further provides a power converter, as described in the following embodiments. Since the principle of solving the problem in the power converter embodiment is similar to that in the above method embodiment, the implementation of the power converter embodiment can refer to the implementation of the above method embodiment, and the repeated parts will not be repeated.

[0060] FIG. 7 shows another schematic structural diagram of a driving circuit of the Boost circuit shown in FIG. 1. The driving circuit includes:

[0061] a first switch **701**, electrically connected to a ground terminal **704** and including a first control terminal;

[0062] a second switch **702**, electrically connected to the first switch **701** to form a common node **703** (Gnd_up), and including a second control terminal;

[0063] a driving power supply **705**, electrically connected to the first switch **701** and the ground terminal **704**;

[0064] a diode **706**, where an anode of the diode **706** is electrically connected to the driving power supply **705**;

[0065] a bootstrap capacitor **707**, electrically connected between the common node **703** and a cathode of the diode **706**;

[0066] a first driving unit **708**, electrically connected to the driving power supply **705**, the first control terminal of the first switch **701** and the ground terminal **704**, and controlling the on and off of the first switch **701**;

[0067] a second driving unit **709**, electrically connected to the bootstrap capacitor **707**, the second control terminal of the second switch **702** and the common node **703**, and controlling the on and off of the second switch **702**; and

[0068] a control unit **710**, electrically connected to the first driving unit **708** and the second driving unit **709**, and configured to:

[0069] when a duty cycle of the first switch **701** decreases to a first threshold, control the duty cycle of the first switch **701** to be a preset duty cycle every time at least one switching cycle elapses, and the preset duty cycle is not less than the first threshold.

[0070] It should be noted that the driving mode of the power converter is bootstrap driving. When the first switch **701** is turned on, the bootstrap capacitor **707**, the first switch **701**, the driving power supply **705** and the diode **706** form a charging loop to charge the bootstrap capacitor **707**.

[0071] The power converter includes the Boost circuit, the totem pole PFC circuit or the three-phase PFC circuit.

[0072] In some embodiments of the present disclosure, the control unit **710** is further configured to:

[0073] when an input voltage of the power converter is positive and the duty cycle of the first switch **701** is controlled to be the preset duty cycle, control the second switch **702** to be turned on after a dead time following the turn-off of the first switch **701**, and the duty cycle of the second switch **702** is the preset duty cycle.

[0074] In some embodiments, the power converter further includes an input power supply, and the polarity of the input voltage may be determined based on the type of the input power supply.

Specifically, when the input power supply is a DC power supply, the input voltage is a DC voltage, and the input voltage of the power converter is always considered positive. When the input power supply is an AC power supply, the input voltage of the power converter is positive during the positive half cycle of the AC power supply.

[0075] In some embodiments, when the power converter is the Boost circuit, the totem pole PFC circuit or the three-phase PFC circuit, the duty cycle of the first switch gradually decreases to zero in a no-load state according to a conventional control strategy. In this case, when the duty cycle of the first switch is reduced to the first threshold, the change process of the duty cycle of the first switch is forcibly changed to make the duty cycle of the first switch become the preset duty cycle, thereby ensuring the charging of the bootstrap capacitor and stabilizing the voltage of the bootstrap capacitor. If the power converter is in a load switching state, for a case where the duty cycle of the first switch gradually decreases, there is a state in which the duty cycle of the first switch is reduced to the first threshold. In this case, if the change process of the duty cycle of the first switch is not forcibly changed, the duty cycle of the first switch may be reduced to below the first threshold (which is not necessarily zero). In this case, the charging of the bootstrap capacitor is not sufficient to drive the normal operation of the second switch, affecting the normal operation of the circuit. For this, it is also necessary to adopt the control method in the present disclosure. When the duty cycle of the first switch is reduced to the first threshold, the duty cycle of the first switch is forcibly changed to become the preset duty cycle, thereby stabilizing the voltage of the bootstrap capacitor and maintaining the normal operation of the circuit.

[0076] In some embodiments, when the power converter is the totem pole PFC circuit, if the input power of the circuit is powered off and has not yet been restored, the conventional control strategy requires reducing the duty cycle of the first switch and the second switch to be reduced to zero. Therefore, when the duty cycle of the first switch is reduced to the first threshold, in order to stabilize the voltage of the bootstrap capacitor, the duty cycle of the first switch is forcibly changed to the preset duty cycle.

[0077] In some embodiments, the control unit **710** is further configured to determine a range of the preset duty cycle, and the minimum value (i.e., the first threshold) of the preset duty cycle is determined by the sum of the energy required to drive the second driving unit **709** and the energy required to turn on the second switch **702**. As shown in FIG. **8**, the maximum value of the preset duty cycle is determined by the input power when the input voltage of the power converter **801** is maximum and the power consumed by the auxiliary power supply **802** electrically connected to the output terminal of the power converter. When the first switch **701** is turned on with the maximum value of the preset duty cycle, where the input voltage of the power converter **801** is at its maximum, the input power of the power converter **801** is less than the power consumed by the auxiliary power supply **802**.

[0078] How the control unit adjusts the duty cycle of the first switch **701** and the second switch **702** and determines the conduction timing of the first switch **701** and the second switch **702** according to the state of the power converter and the direction of the input voltage has been described above, which will not be repeated here.

[0079] Those skilled in the art can understand that various aspects of the present disclosure may be implemented as a system, a method, or a program product. Therefore, various aspects of the present disclosure can be embodied in the following forms: a complete hardware implementation, a complete software implementation (including firmware, microcode, etc.), or a combination of hardware and software implementations, which can be collectively referred to as “circuit”, “module”, or “system”. It should be noted that although several modules or units of devices for executing actions in the above detailed description are mentioned, such division of modules or units is not mandatory. In fact, features and functions of two or more of the modules or units described above may be embodied in one module or unit in accordance with embodiments of the present disclosure. Alternatively, the features and functions of one module or unit described above

may be further divided into multiple modules or units.

[0080] In addition, although various steps of the method of the present disclosure are described in a particular order in the figures, this is not required or implied that the steps must be performed in the specific order, or all the steps shown must be performed to achieve the desired result. Additionally or alternatively, certain steps may be omitted, multiple steps may be combined into one step, and/or one step may be decomposed into multiple steps and so on.

[0081] Through the description of the above embodiments, those skilled in the art will readily understand that the example embodiments described herein may be implemented by software or by a combination of software with necessary hardware. Therefore, the technical solutions according to embodiments of the present disclosure may be embodied in the form of a software product, which may be stored in a non-volatile storage medium (which may be a CD-ROM, a USB flash drive, a mobile hard disk, etc.) or on a network. A number of instructions are included to cause a computing device (which may be a personal computer, server, mobile terminal, or network device, etc.) to perform the methods in accordance with embodiments of the present disclosure.

[0082] Other embodiments of the present disclosure will be apparent to those skilled in the art after those skilled in the art consider the specification and practice the technical solutions disclosed herein. The present application is intended to cover any variations, uses, or adaptations of the present disclosure, which are in accordance with the general principles of the present disclosure and include common general knowledge or conventional technical means in the art that are not disclosed in the present disclosure. The specification and embodiments are illustrative, and the real scope and spirit of the present disclosure is defined by the appended claims.

Claims

1. A method for controlling a power converter, wherein the power converter comprises a first switch, a second switch, a first driving unit corresponding to the first switch and a second driving unit corresponding to the second switch, and a bootstrap capacitor configured to supply power to the second driving unit, when the first switch is turned on, the bootstrap capacitor is charged via the first switch, and wherein the method comprises: when a duty cycle of the first switch is decreased to a first threshold, controlling the duty cycle of the first switch to be a preset duty cycle every time at least one switching cycle elapses, wherein the preset duty cycle is not less than the first threshold.
2. The method according to claim 1, further comprising: when an input voltage of the power converter is positive and the duty cycle of the first switch is controlled to be the preset duty cycle, controlling the second switch to be turned on after a dead time following the turn-off of the first switch.
3. The method according to claim 2, wherein a duty cycle of the second switch is equal to the preset duty cycle.
4. The method according to claim 2, wherein the input voltage of the power converter being positive comprises the input voltage being a direct current voltage or the input voltage being a positive half cycle of an alternating current power supply.
5. The method according to claim 1, wherein when the power converter is in a no-load state, when a load is switched and the duty cycle of the first switch is gradually decreased, or when an input power supply of the power converter is an alternating current power supply that has been powered off and has not yet been restored, a state in which the duty cycle of the first switch is decreased to the first threshold occurs.
6. The method according to claim 1, wherein the preset duty cycle is provided with a minimum value and a maximum value, and the minimum value of the preset duty cycle is equal to the first threshold.
7. The method according to claim 6, further comprising: determining the first threshold based on a

sum of energy required to drive the second driving unit and energy required to turn on the second switch.

8. The method according to claim 6, further comprising: determining the maximum value of the preset duty cycle based on an input power when an input voltage of the power converter is maximum and a power consumed by an auxiliary power supply electrically connected to an output terminal of the power converter, wherein the input power is less than the power consumed by the auxiliary power supply.

9. The method according to claim 1, wherein the power converter comprises a Boost circuit, a totem pole Power Factor Correction (PFC) circuit or a three-phase PFC circuit.

10. A power converter, comprising: a first switch and a second switch, electrically connected with each other to form a common node; a first driving unit, powered by a driving power supply and configured to drive the first switch; a second driving unit, powered by a bootstrap capacitor and configured to drive the second switch; and a control unit, electrically connected to the first driving unit and the second driving unit, and configured to: when a duty cycle of the first switch is decreased to a first threshold, control the duty cycle of the first switch to be a preset duty cycle every time at least one switching cycle elapses, wherein the preset duty cycle is not less than the first threshold.

11. The power converter according to claim 10, further comprising: a diode, wherein an anode of the diode is electrically connected to the driving power supply and the first driving unit, and a cathode of the diode is electrically connected to the bootstrap capacitor and the second driving unit, and when the first switch is turned on, the bootstrap capacitor is charged by the driving power supply via the first switch.

12. The power converter according to claim 10, wherein the control unit is further configured to: when an input voltage of the power converter is positive and the duty cycle of the first switch is controlled to be the preset duty cycle, control the second switch to be turned on after a dead time following the turn-off of the first switch.

13. The power converter according to claim 12, wherein a duty cycle of the second switch is equal to the preset duty cycle.

14. The power converter according to claim 12, wherein the input voltage of the power converter being positive comprises the input voltage being a direct current voltage or the input voltage being a positive half cycle of an alternating current power supply.

15. The power converter according to claim 10, wherein when the power converter is in a no-load state, when a load is switched and the duty cycle of the first switch is gradually decreased, or when an input power supply of the power converter is an alternating current power supply that has been powered off and has not yet been restored, a state in which the duty cycle of the first switch is decreased to the first threshold occurs.

16. The power converter according to claim 10, wherein the preset duty cycle is provided with a minimum value and a maximum value, and the minimum value of the preset duty cycle is equal to the first threshold.

17. The power converter according to claim 10, wherein the control unit is further configured to: determine the first threshold based on a sum of energy required to drive the second driving unit and energy required to turn on the second switch.

18. The power converter according to claim 10, wherein the control unit is further configured to: determine the maximum value of the preset duty cycle based on an input power when an input voltage of the power converter is maximum and a power consumed by an auxiliary power supply electrically connected to an output terminal of the power converter, wherein the input power is less than the power consumed by the auxiliary power supply.

19. The power converter according to claim 10, wherein the power converter comprises a Boost circuit, a totem pole Power Factor Correction (PFC) circuit or a three-phase PFC circuit.
